CSIS 4495 002 Applied Research Project

Progress Report 1

Developing a Diabetes and Prediabetes Prediction Web App Using 2025 CDC BRFSS Data

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Project arrangement: Individual project

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1. Work Log

Date	Hours	Description of work done
Oct 1, 2026	4	Reviewed the 2025 BRFSS codebook, calculated-variable documentation, and related CDC references. Identified candidate variables and compared them with the earlier 2021 diabetes project.
Oct 4, 2026	4	Continued data preparation in Jupyter. Created the three-class diabetes target from DIABETE4, selected _AGE80 for age, created National Report and detailed age groups, and reviewed the sex variable.
Oct 5, 2026	2	Continued background research on the selected BRFSS variables and CDC documentation. Reviewed how calculated variables are defined and how special response codes should be interpreted.
Oct 7, 2026	3	Compared original and calculated BRFSS variables and considered which versions would be more appropriate for the project.
Oct 8, 2026	3	Continued reviewing special response and missing-value definitions and refined the planned treatment of demographic, lifestyle, socioeconomic, and diet variables.
Oct 10, 2026	5	Continued EDA for age and sex. Compared diabetes status across groups, calculated diabetes prevalence for the detailed age groups, created an age-prevalence chart, and added short notes to the notebook.
Oct 11, 2026	6	Worked through the remaining selected predictors: BMI, high blood pressure, smoking, cholesterol, physical activity, education, income, race/ethnicity, fruit and vegetable intake, and general health. Checked the CDC documentation before recoding each variable.
Oct 12, 2026	6	Reviewed missing data and found that complete-case deletion would remove about 40% of the records. Prepared the 13-predictor modeling dataset, completed an 80/20 stratified train/test split, and set up preprocessing for numeric and categorical variables. Reviewed GitHub progress and prepared Progress Report 1, including the work log, summary, repository check-in, AI use, and prompt history.

2. Summary Description of Work

During this reporting period, I moved from the proposal into data preparation and EDA. At the beginning, I spent more time reviewing the 2025 BRFSS codebook, calculated-variable documentation, and related CDC material because many variables use special codes and some calculated variables are more useful than the original survey variables.

I first prepared the diabetes target, age, and sex variables, then gradually worked through the other predictors. The current set includes age, sex, BMI, high blood pressure, smoking, cholesterol, physical activity, education, income, race/ethnicity, fruit and vegetable intake, and general health. I checked the BRFSS documentation before deciding how to recode each variable and how to treat special responses such as “Don’t know” and “Refused.”

The EDA showed several patterns that are worth carrying forward into modeling. Diabetes was more common in older age groups, respondents with higher BMI, and respondents with high blood pressure. Other variables also showed differences, but I am treating these as associations rather than causal relationships. General health especially needs caution because poor general health could be related to diabetes, other chronic conditions, or both.

Missing data became an important issue. Only 59.72% of the records were complete across all selected variables, so I decided not to drop every incomplete row. I prepared a modeling dataset with 13 predictors, used an 80/20 stratified train/test split, and set up preprocessing with separate treatment for numeric and categorical features. The next step is to train and compare Logistic Regression and Random Forest and then continue the Power BI analysis.

The project is currently on schedule with the timeline in my proposal shown as Figure 1. Data preparation was planned for October 1–31, and by October 12 I had completed most of the feature cleaning, EDA, missing-data review, and preprocessing setup. Model development and Power BI work are planned to begin in the second half of October, so the next stage is still consistent with the original schedule.

Work package	Sep 21–30	Oct 1–15	Oct 16–31	Nov 1–15	Nov 16–30	Dec 1–6
Proposal and source audit	●	●				
Data preparation		●	●			
Models and evaluation			●	●		
Power BI visuals			●			
Flask app and SQLite				●	●	
Testing and report					●	●

Figure 1. Original project timeline with completed work highlighted in red.

3. Repository Check-In

I checked the updated Jupyter notebook into the GitHub repository. The repository history shows the work developing in stages: reference and feature review, target and demographic preparation, age/sex EDA, and then the larger feature-cleaning and preprocessing update. The current notebook includes the work completed so far and is organized with section headings so I can continue from the same structure.

GitHub check-ins were less frequent during part of this period because much of the work involved documentation review and variable planning. I returned to more concentrated notebook work near the end of the period and plan to make smaller check-ins as the modeling work continues.

4. AI Use

AI tool, version and account	Specific use	Student contribution beyond AI
ChatGPT, GPT-5.6 Sol, personal ChatGPT account	Used to discuss feature choices, data-cleaning decisions, preprocessing options, notebook organization, and wording of short EDA notes.	I made the final variable and method decisions, checked definitions against the CDC documentation, ran the notebook, reviewed the outputs, and decided how to interpret the results.

Appendix A. AI Prompt History

The following are the main project questions I discussed with ChatGPT during this reporting period.

Prompt	Question / topic discussed
1	Should heart attack, coronary heart disease, and stroke be used as diabetes risk factors, or could they be outcomes/complications that are not appropriate for an early prediction model?
2	Should use the National Diabetes Statistics Report age groups (18–44, 45–64, and 65+) or more detailed age groups for the EDA?
3	Should use the original BRFSS variable or the CDC calculated variable for analysis?
4	Should “Don’t know” and “Refused” be treated as missing values based on the BRFSS codebook?
5	For BMI, should extreme values automatically be removed as outliers, or should they be kept?
6	Which smoking categories should be kept, and is it useful to keep former smokers separate from current and never smokers?
7	How should differences in diabetes prevalence across race/ethnicity groups be interpreted?
8	Is it useful to compare race/ethnicity with physical activity and diet?
9	How should the relationship between General Health and diabetes be interpreted?
10	Prediabetes is only about 2.9% of the sample. When building the model, what should I pay attention to?

Oct 1 — Prompt 1

About the relationship among heart attack, coronary heart disease, stroke and diabetes, should them to be used as diabetes risk factors, or could they be ooutcomes that are not good for an early prediction model?

Yes. For an **early prediction model**, I would be careful about using heart attack, coronary heart disease, and stroke as predictors.

These conditions are strongly related to diabetes, but diabetes itself can increase the risk of cardiovascular disease and stroke. They also share common factors such as age, obesity, high blood pressure, and high cholesterol. Because of this, it may be difficult to know whether these conditions occurred before or after diabetes.

If the goal is to predict diabetes **before diagnosis**, including them could make the model depend on conditions that may already be consequences or complications of diabetes.

I would therefore exclude them from the main early-prediction model and focus on variables that are more appropriate for earlier risk assessment, such as age, BMI, blood pressure, cholesterol, smoking, physical activity, diet, education, and income.

They can still be discussed in the report as **related health conditions**, but I would avoid describing them as simple causes of diabetes.

Oct 2 — Prompt 2

Should I use the National Diabetes Statistics Report age groups (18–44, 45–64, and 65+) or more detailed age groups for the EDA?

I would use **both**, but for different purposes.

For comparison with the National Diabetes Statistics Report, use the broader groups **18–44, 45–64, and 65+**. This makes your results easier to compare with the national report.

For your own EDA, the more detailed age groups are more useful because they show how diabetes prevalence changes more gradually with age. For example, they can show differences between people in their 50s, 60s, and 70s that would be hidden inside one large group.

So I would recommend:

- **National age groups** for external comparison.
- **Detailed age groups** for deeper EDA.
- **Continuous age** can still be used later in the prediction model instead of forcing age into groups.

This gives you both comparability and more detailed insight.

Oct 6 — Prompt 3

When BRFSS provides both an original survey variable and a calculated variable for the same topic, which one is better to use for analysis? Should I prefer the CDC calculated variable when it is already cleaned or grouped for analysis?

Usually, I would prefer the **CDC calculated variable** when it is clearly designed for analysis and its definition matches your research purpose.

Calculated variables are often useful because CDC has already combined or standardized the original responses into a cleaner form. This can make categories more consistent and easier to compare.

However, I would not automatically choose the calculated variable every time. I would first check:

- what the calculated variable represents,
- how CDC created it,
- whether any information was lost through grouping,
- and whether the original variable gives more detail that is useful for your project.

So my approach would be: **use the calculated variable when it is appropriate for the analysis, but check the codebook first and compare it with the original variable before deciding.**

Oct 8 — Prompt 4

Many BRFSS variables use special codes such as "Don't know" or "Refused." Should these values be treated as missing instead of keeping them as normal categories? Should I decide this based on the codebook for each variable?

Yes. In most cases, values such as **"Don't know"** and **"Refused"** should not be treated as normal response categories.

They usually mean that the actual value is unknown, so I would normally treat them as **missing values** for analysis.

However, I would still check the **BRFSS codebook for each variable**, because the special codes are not always the same across variables. Some variables may also include codes such as "Not asked" or "Not applicable," which can have a different meaning from "Don't know" or "Refused."

So I would use this rule:

Check the codebook first, then decide how each special code should be handled. Do not apply one general rule to every BRFSS variable.

Oct 10 — Prompt 5

For BMI, should extreme values automatically be removed as outliers, or should they be kept ?

I would **not automatically remove extreme BMI values** just because they look like outliers.

First, check whether the values are valid according to the BRFSS documentation and whether they are plausible. Some very high or very low BMI values may represent real respondents, not data errors.

For your project, I would suggest:

- Keep extreme BMI values if they are valid and plausible.
- Remove or correct them only if there is clear evidence that they are invalid.
- Check how much they affect the model, especially Logistic Regression.
- You can compare results with and without extreme values as a sensitivity check if needed.

This is especially important because Random Forest is usually less sensitive to extreme values than Logistic Regression.

So I would not use an automatic IQR rule to delete BMI outliers. I would base the decision on the data definition, plausibility, and model impact.

Oct 10 — Prompt 6

Which smoking categories should be kept, and is it useful to keep former smokers separate from current and never smokers?

I would keep **former smokers separate** from current and never smokers.

For your BRFSS variable, the four useful categories are:

- Current smoker — every day
- Current smoker — some days
- Former smoker
- Never smoked

Former smokers are important because they may still have health effects from past smoking, and some people may have stopped smoking because of health problems. Combining them with never smokers would lose that information.

For EDA, I would keep all four categories. For modeling, you can also keep them as separate categorical levels and use one-hot encoding.

If you later want a simpler model, you could compare it with a version that combines the two current-smoking categories, but I would not combine former smokers with never smokers.

Oct 11 — Prompt 7

For Race/Ethnicity, some groups have much smaller sample sizes than others. For example, the Asian group is only about 2.8% of the sample, although it still includes around 10,000 respondents. I do not think differences in diabetes prevalence should be explained by race alone. They may also be related to differences in age, BMI, income, education, diet, physical activity, and other factors. Do you think this is a reasonable way to interpret the results?

Yes, I think this is a reasonable interpretation.

The sample-size difference is important, but it is not the only issue. Even if the Asian group has around 10,000 respondents, the groups may have very different characteristics. For example, one racial group may be younger on average, have different BMI levels, different income or education levels, or different lifestyle patterns.

Because of this, the diabetes prevalence for each race/ethnicity group should be treated as a **descriptive comparison**, not as evidence that race itself causes the difference.

A good next step would be to compare race/ethnicity with other variables such as:

- age
- BMI
- physical activity
- smoking
- income
- education
- fruit and vegetable intake

This can help show whether the observed diabetes differences may partly reflect other characteristics of each group.

Oct 11 — Prompt 8

Based on the Race/Ethnicity results, I am thinking about comparing race/ethnicity with physical activity and diet, such as fruit and vegetable intake. This may help me understand whether some of the differences in diabetes prevalence between racial groups are also related to lifestyle differences. Do you think this is a useful analysis to include in my report?

Yes, I think this would be a useful analysis to include.

Comparing race/ethnicity with physical activity and diet can help you see whether the groups also differ in lifestyle patterns. This may provide more context for the differences you observed in diabetes prevalence.

I would suggest using **percentages within each racial/ethnic group** rather than raw counts, because the group sizes are very different.

You should still describe the results as **associations**, not causes. If one group has lower physical activity or lower fruit and vegetable intake, that does not mean race itself causes those behaviors or the diabetes difference.

For the report, I think these comparisons would be enough:

- Race/Ethnicity × Physical Activity
- Race/Ethnicity × Fruit Intake
- Race/Ethnicity × Vegetable Intake

If these results show clear patterns, you can discuss them briefly as possible lifestyle differences that may help explain part of the variation between groups.

Oct 11 — Prompt 9

About General Health, I do not think it should be treated simply as a factor that causes diabetes, and I also do not think it should be treated only as a result of diabetes. I think the relationship may be interdependent, because poor general health can be related to diabetes, while diabetes can also affect a person's overall health. When analyzing this variable, I think we need to consider this two-way relationship. Do you think this interpretation is reasonable?

Yes, I think your interpretation is reasonable.

General Health should not be interpreted as a simple one-way causal factor. Poor general health may be associated with diabetes because of other conditions, lifestyle factors, age, or obesity, while diabetes itself can also reduce a person's overall health. Therefore, the relationship can be bidirectional and may also be influenced by shared underlying factors.

For your project, I would treat General Health mainly as a **predictive variable**, not as a clear causal risk factor. In the report, you should describe the relationship as an **association**.

A good way to handle it is to compare two models later:

- one model **with General Health**
- one model **without General Health**

If the model performs much better with it, that shows it has predictive value. But you should still avoid saying that poor general health causes diabetes.

Oct 12 — Prompt 10

Prediabetes is only about 2.9% of the sample, when build model, what shall be pay attention to?

Because Prediabetes is only about **2.9% of the sample**, class imbalance will be an important issue.

When building the model, I would pay attention to these points:

- Do not rely on **accuracy alone**, because a model could have high accuracy while predicting very few Prediabetes cases correctly.
- Use **precision, recall, and F1-score for each class**, especially the Prediabetes class.
- Use a **stratified train/test split** so the class proportions stay similar in both sets.
- Compare methods such as **class weights** and possibly **SMOTE** to see whether they improve Prediabetes prediction.
- If SMOTE is used, apply it **only to the training data**, not the test data.
- Check the **confusion matrix** carefully to see whether Prediabetes cases are being classified as No Diabetes or Diabetes.
- Also report **macro F1** in addition to weighted F1, because macro F1 gives the minority class more equal importance.

For this project, I think the **Prediabetes F1-score and recall** should be especially important, since identifying this small group is one of the main goals of the model.